

MOHAMED MARTINI

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SUMMARY

MLE with proven industry skills in owning the full cycle of ML solutions, cloud-native backend applications and microservices design. Passionate about architecting scalable, reliable, and maintainable ML and data intensive applications that leverage AI and timeless engineering principles.

SKILLS

- **Programming Languages:** Expert in Python, SQL. Familiar with C/C++, Dart.
- **AI/ML:** TensorFlow, PyTorch, Pandas, LangChain, LangGraph, RAG, GCV, HuggingFace, Agentic Workflows (Claude).
- **Data Engineering:** Apache Beam, BigQuery, Postgres, GCS, Weaviate, Kafka, Pub/Sub, PySpark, Airflow, ETL.
- **Backend & Architecture:** FastAPI, REST, gRPC, WebSockets, SSE, Webhooks, Auth/Authz.
- **Containerization & CI/CD:** Docker, Docker Compose, GitHub Actions, Python workspace management.

PROFESSIONAL EXPERIENCE

Lever AI

Jul 2025 - Present

Founder & Principal Engineer

- Founded a consulting and SaaS development LLC specialized in AI-powered, data-driven software solutions. **Projects:**
 - Mobile numerical estimation game published to [Android](#) and [iOS](#) featuring: **LLM-native ETL pipeline** to generate a database of >1k semantically unique and high-quality estimation questions; **Cross-Service Observability** via end-to-end distributed tracing; **High-consistency and low-latency** multiplayer game mode; **GitFlow CI/CD** pipelines with automated unit, integration, and smoke testing, and dev/prod deployments. Leveraged **AI coding agents** (Claude, Gemini) to accelerate development while maintaining full architectural oversight.
 - **Stack:** FastAPI, LangChain, Firestore, Firebase, Postgres with pgvector, OpenTelemetry, Flutter.
 - ThermoPulse (IP): Industrial cold-chain monitoring system with time-series analytics and early fault detection.
 - Sanad (IP): RAG and temporal citator tracking amendments and repeals of Syrian law from 300k pdf scans.

Pison Technology

Oct 2022 - Jul 2025 (2 years, 10 months)

Machine Learning Engineer III

- Promoted from MLE to MLE III in April 2025.
- Owned the full cycle of a sleep-stage classification model. **Improved the baseline macro F1-score by 15%.** Ran 1,200+ TensorFlow experiments for sleep stage classification. **Mitigated class imbalance** with Cost-Sensitive and Contrastive Losses. Designed a **custom gating layer for feature mining**, improving performance while reducing the size of the feature set by an order of magnitude. **Monitored SPI** of predictions to trigger re-training alerts on data drift.
- Developed real-time biosignal gesture detection models. Wrote Cythonized extension modules, and vectorized bottleneck functions, significantly improving the speed of online prediction and offline modeling.
- Led a “Hackaweek” project with two colleagues to implement a SOTA **domain-invariant classification network** with adversarial training in PyTorch, following the approach outlined in its original manuscript.
- Led the adoption of **BigQuery** for analytics pipelines. Decided **partitioning and clustering** according to query patterns.
- Deployed and monitored three services using Google **Cloud Run** and **Cloud Functions**.
- **Improved the speed of a batch ETL pipeline** using Apache Beam by more than **190 folds**.
- Combined standalone Python repos into a single uv workspace, allowing teams on different operating systems to seamlessly cooperate, reducing the size of deployed containers, and streamlining code installation, testing, and deployment.
- Maintained cross-team communication to ensure consistency and optimal integration of ML solutions.

University of Massachusetts Lowell

Jan 2021 – June 2022 (1 year, 1 month)

Research Assistant

- Implemented **Reinforcement Learning** agents and environments for cooperative multi-agent Search and Rescue tasks.
- Implemented **object detection** training and inference pipelines. Used Generative AI for data augmentation.

EDUCATION

University of Massachusetts Lowell

- *Master's in Computer Engineering* | GPA 4.0 2021-2022
- *Bachelor's in Electrical Engineering* | GPA 3.8 2018-2021